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Devices and systems for flying bitline with jumper cell

Abstract

The disclosed device can include a bitcell array located on a first metal layer including a first subarray of bitcells and a second subarray of bitcells; a first write driver device coupled to the first subarray of bitcells from a first end of the first subarray; a second write driver device coupled to the second subarray of bitcells from a first end of the second subarray; a third write driver device coupled to the first subarray of bitcells from a second end of the first subarray; and a fourth write driver device coupled to the second subarray of bitcells from the second end of the second subarray. Various other devices, systems, and methods of manufacture are also disclosed.

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Background/Summary

BACKGROUND

(1) Smaller and more powerful computing devices continue to be in demand. To fit more memory on a chip, the density of memory circuits may be increased. Increasing the density of memory circuits may, in some Static Random Access Memory (SRAM) designs, involve longer bitlines. However, longer bitlines may be more resistive, which may degrade the quality of the signal on the bitline for writing to the bitcell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate a number of example embodiments and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.

(2) FIG. 1 is a diagram of an example bitcell array design.

(3) FIG. 2 is a illustration of bitcell performance under differing conditions.

(4) FIG. 3 is an illustration of various example bitcell array designs.

(5) FIG. 4 is an illustration of an example flying bitline with jumper cell design.

(6) FIG. 5 is a timing diagram of an example flying bitline with jumper cell.

(7) FIG. 6 is an illustration of bitline performance under differing conditions.

(8) FIG. 7 is an illustration of an example jumper cell design.

- (9) FIG. 8 is an additional illustration of an example jumper cell design.
- (10) FIG. 9 is a diagram of an example floorplan for a memory array.
- (11) FIG. 10 is a flow diagram of an example method of manufacture for an example flying bitline with jumper cell.
- (12) Throughout the drawings, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the example embodiments described herein are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. However, the example embodiments described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within the scope of the appended claims.

DETAILED DESCRIPTION OF EXAMPLE IMPLEMENTATIONS

- (13) The present disclosure is generally directed to devices and systems for flying bitlines with jumper cells. The flying bitline described herein may enable designs that improve density and write speed. In addition, the devices and systems described herein may incorporate write drivers on both ends of the flying bitline, thereby potentially further improving write speed. A “bottom” bitline may extend from one write driver to the other, as may a “top” bitline, each serving half of the bitcells. The bottom bitline may jump at the jumper cell from one metal layer (e.g., M0) to another (e.g., M2). The top bitline may do the same, but with the positions of the M0 portion and the M2 portion switched. The jumper cell design may allow jumps on both sides by using a two-step “jog” jump pattern, enabling the flying bitline with two write drivers design.
- (14) The devices and systems described herein may contribute to increased memory density, improved reliability and writeability of bitcells, and/or improved write speed.
- (15) FIG. 1 is a diagram of an example bitcell array device **100**. In one example, bitcell array device **100** may form part of an SRAM architecture. As shown in FIG. 1, device **100** may include a bitcell array **102** and a bitcell array **104**. Bitcell array **102** may include a bitline (“BL <0>”) and a complementary bitline (“BLB <0>”), each of which may run the length of bitcell array **102**, connecting various individual bitcells. Bitcell array **104** may similarly include a bitline BL <1> and a complementary bitline BLB <1>.
- (16) A write driver **110** may include a Metal-Oxide-Semiconductor-Field-Effect-Transistor (“MOSFET”), such as an n-channel MOSFET (“NMOS”), that connects to BL <0>, where a write control signal “WRCS <0>” connects to the gate of the MOSFET and a DATA signal connects to the source of the MOSFET. Thus, write driver **110** may amplify the DATA signal and, when WRCS <0> is active, the MOSFET may send the DATA signal to BL <0>. Similarly, an NMOS may connect to BLB <0>, also gated by WRCS <0> and sourced by a complementary DATA_X signal. Write driver **110** may perform parallel functions for bitcell array **104**, with a write control signal WRCS <1> gates the flow of the DATA signal to BL <1> and gates the flow of the DATA_X signal to BLB <1>.
- (17) As used herein, the term “bitcell” may refer to any circuit that stores a bit of data. For example, the term “bitcell” may refer to a circuit that stores a single bit of data in an SRAM devices. In some examples, a bitcell may include two cross-coupled inverters and two access transistors that connect the bitcell to one or more bitlines (e.g., a bitline and a complementary bitline) and a wordline.
- (18) As used herein, the term “write driver” may refer to any circuit that is controlled by one or more write control signals to write data to selected bitcells. In some examples, a write driver may amplify an input data signal to a selected bitline to write data to a selected bitcell.
- (19) FIG. 2 illustrates a graph **200** showing bitcell performance under differing conditions. Thus, for example, graph **200** shows a waveform BL_High_Res <0> showing the signal on a higher-resistance bitline and a waveform BL_Low_Res <0> showing a signal on lower-resistance bitline. WRCS shows a write control signal and WL shows a word line signal. TR shows a “true” signal of

an internal node of a bitcell and FL shows a complementary signal of an internal node of a bitcell, where the bitcell is on the lower-resistance bitline. TR' and FL' show the true and complementary signals where the bitcell is on the higher-resistance bitline. As can be seen, the write time of a bitcell may be decreased by decreasing the resistance of the bitline.

(20) FIG. 3 is an illustration of various example bitcell array devices **302**, **304**, and **306**. As shown in FIG. 3, bitcell array device **302** may be similar to device **100**. In one example, device **302** may represent a bitcell array with 256 bits per bitline.

(21) Bitcell array device **304** may be similar to bitcell array device **302**, except bitcell array device **304** may have twice as many bitcells (e.g., 512 bits per bitline). Because the layout, structure, and proportions of device **304** may otherwise be similar to device **302**, to fit 512 bits per bitline, the bitlines of device **304** may be approximately twice as long as those of device **302**. The increased length of the bitlines may result in a higher bitline resistance to BIT <0> at the far end of the bitline from the write driver, potentially resulting in slower write times.

(22) Bitcell array device **306** may divide a bitcell array into bitcell subarrays **310** and **312**.

Although depicted side-by-side for illustrative purposes, bitcell subarray **312** may be in line with bitcell subarray **310**. In one example, bitcell subarrays **310** and **312** may both run on the M0 metal layer. A separate bitline **314** on the M2 metal layer (e.g., running underneath bitcell subarray **310**) may connect bitcell subarray **312** to the write driver.

(23) Bitcell subarray **310** may have a similar structure, layout, and speed to device **302**. Bitcell subarray **312** may be slower than bitcell subarray **310**. Thus, device **306** may be more dense than but slower than device **302**, and less dense but faster than device **304**.

(24) FIG. 4 is an illustration of an example flying bitline with jumper cell device **400**. As shown in FIG. 4, device **400** may include a bitcell subarray **402** and a bitcell subarray **404**, whose proximate ends meet at a jumper cell **422** and whose distal ends reach write driver **410** and write driver **420**. Thus, device **400** may include a write driver **410** on one end of bitcell subarray **402** and bitcell subarray **404** and a write driver **420** on the other end of bitcell subarray **402** and bitcell subarray **404**.

(25) As used herein, the term “bitcell subarray” may generally refer to a portion of a bitcell array. In some examples, the term “bitcell subarray” may refer to a portion of a bitcell array that is at least partially separate from another portion of the bitcell array. For example, two bitcell subarrays in the same array may be separated by a jumper cell. Additionally or alternatively, two such bitcell subarrays may not be regulated by the same write driver components and/or signals (e.g., the same MOSFETs, the same write control signals, etc.). Nevertheless, in various examples two such bitcell subarrays may be regarded as part of the same bitcell array. For example, two such bitcell arrays may be in a single line (e.g., with a slight interruption at a jumper cell), and, thus, may be laid out in a similar fashion to a single bitcell array. Additionally, or alternatively, the bitcells of two such bitcell arrays may be addressed as part of a single bitcell array.

(26) While the foregoing discussion focuses on the “true” bitlines **430**, **434**, **440**, and **444**, it can be appreciated that parallel considerations apply to the corresponding complementary bitlines **432**, **436**, **442**, and **446**. In some examples, a bitline **430** (or “first bitline”) may run along bitcell subarray **402**. Bitline **430** may connect to a write driver **410**. A write control signal WRCS_BOT_M0<0> may gate the flow of a DATA signal at a MOSFET to bitline **430**. In addition, in some examples, a bitline precharge device may precharge bitline **430** in response to a BLPCH signal.

(27) In addition, bitline **430** may be extended by bitline **434** to write driver **420**. Bitlines **430** and **434** may run on different metal levels. For example, bitline **430** may run on M0 and bitline **434** may run on M2. While sometimes described herein separately (including, e.g., being referred to as “first” and “third” bitlines), bitlines **430** and **434** may be understood as a single bitline, albeit running on different metal levels. For example, bitlines **430** and **434** may connect via jumper cell **422**. Because the bitline composed of bitlines **430** and **434** may be sent a DATA signal (and, in

some examples, precharged) from both sides of bitcell subarray **402** (e.g., from write driver **410** and write driver **420**), write operations on bitcell subarray **402** may be performed more quickly and/or reliably.

(28) In a similar fashion, a bitline **440** (or “second bitline”) may run along bitcell subarray **404**. Bitline **440** may connect to write driver **420**. A write control signal WRCS_TOP_M0<0> may gate the flow of a DATA signal at a MOSFET to bitline **440**. In addition, in some examples, a bitline precharge device may precharge bitline **440** in response to a BLPCH signal.

(29) In addition, bitline **440** may be extended by bitline **444** to write driver **410**. Bitlines **440** and **444** may run on different metal levels. For example, bitline **440** may run on M0 and bitline **444** may run on M2. While sometimes described herein separately (including, e.g., being referred to as “second” and “fourth” bitlines), bitlines **440** and **444** may be understood as a single bitline, albeit running on different metal levels. For example, bitlines **440** and **444** may connect via jumper cell **422**. Because the bitline composed of bitlines **440** and **444** may be sent a DATA signal (and, in some examples, precharged) from both sides of bitcell subarray **404** (e.g., from write driver **410** and write driver **420**), write operations on bitcell subarray **404** may be performed more quickly and/or reliably.

(30) While depicted side-by-side for easier visualization, bitcell subarray **402** and bitcell subarray **404** may be located such that they run along the same line (e.g., being substantially aligned). For example, bitline **430** may run over bitline **444** (being substantially vertically aligned with each other), and bitline **440** may run over bitline **434** (being substantially vertically aligned with each other).

(31) FIG. 5 shows timing diagrams **510** and **520** of an example flying bitline with jumper cell (e.g., the device **400** shown in FIG. 4). As shown in timing diagrams **510** and **520**, a data signal may activate. A bitline precharge signal may activate, and write control signals from both sides of each composite bitline (e.g., WRCS_BOT_M0 and WRCS_BOT_M2 for the composite bitline of one subarray, and WRCS_TOP_M0 and WRCS_TOP_M2 for the composite bitline of the other subarray) may activate. The corresponding bitline signal for the corresponding composite bitline (BL_BOT/BL_TOP) may then activate. The WL signal may turn on, and then the bitcell may be written.

(32) FIG. 6 shows a graph **600** of bitline performance under differing conditions. As shown in FIG. 6, in a case 3 (e.g., corresponding to device **306** of FIG. 3) may represent the slowest write operation of the three depicted cases. A case 2 (e.g., corresponding to device **304** of FIG. 3) may represent a faster write operation than case 3. A case 1 (e.g., corresponding to device **400** of FIG. 4, may represent the fastest write operation of the three depicted cases. Significantly, case 1 may also provide nearly twice the bit density of case 2.

(33) FIG. 7 is an illustration of an example jumper cell device **700**. As shown in FIG. 7, device **700** may include components on multiple metal levels (e.g., M0, M1, and M2). While M0, M1, and M2 are used by way of example, it may be appreciated that other combinations and/or arrangements of metal levels may be used. For example, in place of M1, device **700** may use M3.

(34) Device **700** may include a bitline set **702** and a bitline set **704**. For example, bitline set **702** may correspond to “true” bitlines, and bitline set **704** may correspond to corresponding complementary bitlines. Thus, it may be appreciated that bitline set **704** may have a parallel and/or symmetrical design to bitline set **702** (thus, for example, bitline set **704** is depicted as a version of bitline set **702** but rotated 180 degrees).

(35) As shown in FIG. 7, device **700** may include a bitline **710** running on M0. In some examples, bitline **710** may correspond to bitline **430** of device **400** in FIG. 4 and/or the “first bitline” referred to herein. Device **700** may also include a bitline **712** running on M0. In some examples, bitline **710** may correspond to bitline **440** of device **400** in FIG. 4 and/or the “second bitline” referred to herein. Device **700** may also include a bitline **716** running on M2. In some examples, bitline **716** may correspond to bitline **434** of device **400** in FIG. 4 and/or the “third bitline” referred to herein.

Device **700** may also include a bitline **714** running on M2. In some examples, bitline **714** may correspond to bitline **444** of device **400** in FIG. **4** and/or the “fourth bitline” referred to herein.

(36) In some examples, bitline **712** may connect to bitline **714** by way of a connection element **718**. For example, one or more vias may connect bitline **712** to bitline **714**, either directly and/or via a connection element **718** (e.g., on M1). As can be seen, the connection of bitlines **712** and **714** may not require any deviation from the alignment of bitlines **712** and **714**. In contrast, bitline **710** may connect to bitline **716** by way of a series of connection elements (e.g., connection element **720**, connection element **722**, and connection element **724**) that form a connection path that may deviate from the line along which bitlines **710**, **712**, **714**, and **716** run.

(37) The architecture of jumper cell device **700** may allow the two composite bitlines (bitline **712** together with bitline **714**, and bitline **710** together with bitline **716**) to both extend in both directions (e.g., to write drivers at either end of the bitcell array). This may enable faster and/or more reliable write operations with a dense memory circuit configuration.

(38) FIG. **8** is an illustration of an example memory device **800** with a jumper cell **802**. As shown in FIG. **8**, jumper cell **802** may have a similar structure to that depicted by device **700** in FIG. **7**, and may enable BL_TOP (e.g., a bitline for a top half of a bitcell array) and BL_BOT (e.g., a bitline for a bottom half of the same bitcell array), extend in both directions (to the left and to the right, as depicted in FIG. **8**), while occupying the same area over two metal levels. Equivalent features apply to the complementary bitlines BLB_TOP and BLB_BOT.

(39) FIG. **9** is an example floorplan for a memory array **900**. As shown in FIG. **9**, memory array **900** may include a bank **902** and a bank **904**. In one example, banks **902** and **904** may each correspond to device **700**. In bank **902**, a bitline BL0 may connect to a read/write multiplexer circuit **910** on one end (at M2) and a local read/write block **912** on the other end (at M0). Similarly, a bitline BL1 may connect to read/write multiplexer circuit **910** on one end (at M0) and local read/write block **912** on the other end (at M2). Likewise, in bank **904**, a bitline BL2 may connect to a read/write multiplexer circuit **914** on one end (at M0) and local read/write block **912** on the other end (at M2). Similarly, a bitline BL3 may connect to read/write multiplexer circuit **910** on one end (at M2) and local read/write block **912** on the other end (at M0). While FIG. **9** shows four sets of 256 rows, systems and devices described herein (such as memory array **900**) may include any suitable number of rows per set and may include any suitable number of sets of rows. For example, the overall number of rows (e.g., owing to the number of rows per set and/or to the number of sets) may be greater than 1,024. In addition, while FIG. **9** shows arrays of equal sizes, in some examples the sizes of the arrays shown in FIG. **9** may vary. For example, the sizes of the arrays proximate to local read/write block **912** may be larger or smaller than the sizes of the arrays further from local read/write block **912**.

(40) FIG. **10** is a flow diagram of an example method **1000** of manufacture for a device with a flying bitline and jumper cell. As shown in FIG. **10**, at step **1010** method **1000** may include disposing, in a first metal layer, a bitcell array including a first subarray of bitcells and a second subarray of bitcells. Step **1010** may also include disposing in the first metal layer a first bitline running on the first metal layer that connects the first subarray of bitcells together and connects to a first write driver device. Step **1010** may further include disposing in the first metal layer a second bitline that connects the second subarray of bitcells together and connects to a second write driver device.

(41) At step **1020**, method **1000** may include disposing, in a second metal layer, a third bitline that connects the first subarray of bitcells to the third write driver and a fourth bitline that connects the second subarray of bitcells to the fourth write driver.

(42) In some examples, a method of manufacture that includes steps **1010** and **1020** may also include disposing a jumper cell between the first subarray of bitcells and the second subarray of bitcells, such that the first bitline and the third bitline are connected by the jumper cell and the second bitline and the fourth bitline are connected by the jumper cell.

(43) In further examples, a method of manufacture may include manufacturing a jumper cell by connecting a first bitline and a third bitline by a via running from the first metal layer to the second metal layer. The method of manufacturing the jumper cell may also include connecting a second bitline to a fourth bitline by a series of connectors that are connected by vias running between metal layers and that run out of line with the first, second, third, and fourth bitlines by running laterally away from the second bitline, running in the direction of the first, second, third, and fourth bitlines, and then running laterally toward the fourth bitline.

(44) The process parameters and sequence of steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein can be shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various example methods described and/or illustrated herein can also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

(45) The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the example implementations disclosed herein. This example description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The implementations disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to the appended claims and their equivalents in determining the scope of the present disclosure.

(46) Unless otherwise noted, the terms “connected to” and “coupled to” (and their derivatives), as used in the specification and claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms “a” or “an,” as used in the specification and claims, are to be construed as meaning “at least one of.” Finally, for ease of use, the terms “including” and “having” (and their derivatives), as used in the specification and claims, are interchangeable with and have the same meaning as the word “comprising.”

Claims

1. A device comprising: a bitcell array located on a first metal layer comprising: a first subarray of bitcells; and a second subarray of bitcells; at least one jumper cell, wherein the first subarray of bitcells and the second subarray of bitcells are disposed on opposite sides of the at least one jumper cell; a first write driver device coupled, via the at least one jumper cell, to the first subarray of bitcells from a first end of the first subarray; a second write driver device coupled, via the at least one jumper cell, to the second subarray of bitcells from a first end of the second subarray; a third write driver device coupled, via the at least one jumper cell, to the first subarray of bitcells from a second end of the first subarray; and a fourth write driver device coupled, via the at least one jumper cell, to the second subarray of bitcells from the second end of the second subarray.
2. The device of claim 1, wherein the second end of the first subarray and the second end of the second subarray are proximate ends and the first end of the first subarray and the first end of the second subarray are distal ends.
3. The device of claim 1, further comprising: a first bitline running on the first metal layer that connects the first subarray of bitcells together and connects to the first write driver device; a second bitline running on the first metal layer that connects the second subarray of bitcells together and connects to the second write driver device; a third bitline running on a second metal layer that connects first subarray of bitcells to the third write driver device; and a fourth bitline running on the second metal layer that connects the second subarray of bitcells to the fourth write driver device.
4. The device of claim 3, wherein: the first and second write driver devices are located on the first metal layer; and the third and fourth write driver devices are located on the second metal layer.

5. The device of claim 3, wherein: the first bitline connects to the third bitline via the at least one jumper cell; and the second bitline connects to the fourth bitline via the at least one jumper cell.
6. The device of claim 5, wherein the jumper cell comprises: a first via connecting the first bitline to the third bitline; a second via connecting the second bitline to a connection path that runs at least in part on a third metal layer; and a third via connecting the connection path to the fourth bitline.
7. The device of claim 3, wherein: the first bitline is vertically aligned over the fourth bitline; and the second bitline is vertically aligned over the third bitline.
8. The device of claim 3, further comprising: a first precharge device that precharges the first bitline; a second precharge device that precharges the second bitline; a third precharge device that precharges the third bitline; and a fourth precharge device that precharges the fourth bitline.
9. The device of claim 3, further comprising a set of complementary bitlines arranged in parallel with the first, second, third, and fourth bitlines.
10. The device of claim 1, wherein the first and second subarrays are substantially in line.
11. A system comprising: a memory circuit comprising: a series of bitcell arrays, the series of bitcell arrays comprising a bitcell array located on a first metal layer comprising: a first subarray of bitcells; and a second subarray of bitcells; at least one jumper cell, wherein the first subarray of bitcells and the second subarray of bitcells are disposed on opposite sides of the at least one jumper cell; a first write driver device coupled, via the at least one jumper cell, to the first subarray of bitcells from a first end of the first subarray; a second write driver device coupled, via the at least one jumper cell, to the second subarray of bitcells from a first end of the second subarray; a third write driver device coupled, via the at least one jumper cell, to the first subarray of bitcells from a second end of the first subarray; and a fourth write driver device coupled, via the at least one jumper cell, to the second subarray of bitcells from the second end of the second subarray; and a processing device that is communicatively coupled to the memory circuit and that is configured to: store data in the memory circuit; and process data retrieved from the memory circuit.
12. The system of claim 11, wherein the second end of the first subarray and the second end of the second subarray are proximate ends and the first end of the first subarray and the first end of the second subarray are distal ends.
13. The system of claim 11, further comprising: a first bitline running on the first metal layer that connects the first subarray of bitcells together and connects to the first write driver device; a second bitline running on the first metal layer that connects the second subarray of bitcells together and connects to the second write driver device; a third bitline running on a second metal layer that connects first subarray of bitcells to the third write driver device; and a fourth bitline running on the second metal layer that connects the second subarray of bitcells to the fourth write driver device.
14. The system of claim 13, wherein: the first and second write driver devices are located on the first metal layer; and the third and fourth write driver devices are located on the second metal layer.
15. The system of claim 13, wherein: the first bitline connects to the third bitline via the at least one jumper cell; and the second bitline connects to the fourth bitline via the at least one jumper cell.
16. The system of claim 15, wherein the jumper cell comprises: a first via connecting the first bitline to the third bitline; a second via connecting the second bitline to a connection path that runs at least in part on a third metal layer; and a third via connecting the connection path to the fourth bitline.
17. The system of claim 13, wherein: the first bitline is vertically aligned over the fourth bitline; and the second bitline is vertically aligned over the third bitline.
18. The system of claim 13, further comprising: a first precharge device that precharges the first bitline; a second precharge device that precharges the second bitline; a third precharge device that precharges the third bitline; and a fourth precharge device that precharges the fourth bitline.
19. The system of claim 13, further comprising a set of complementary bitlines arranged in parallel with the first, second, third, and fourth bitlines.

20. A method of manufacture comprising: disposing, in a first metal layer: a bitcell array comprising: a first subarray of bitcells; and a second subarray of bitcells; at least one jumper cell, wherein the first subarray of bitcells and the second subarray of bitcells are disposed on opposite sides of the at least one jumper cell; a first bitline that connects, via the at least one jumper cell, the first subarray of bitcells together and connects to a first write driver device; and a second bitline that connects, via the at least one jumper cell, the second subarray of bitcells together and connects to a second write driver device; and disposing, in a second metal layer: a third bitline that connects, via the at least one jumper cell, first subarray of bitcells to the third write driver device; and a fourth bitline that connects, via the at least one jumper cell, the second subarray of bitcells to the fourth write driver device.
